PAPER ID—16014

B. Tech. EXAMINATION, 2024

(Third Semester)

ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

Advanced Data Structure

Time : 3 *Hours* *Maximum Marks* : 70

Before answering the question-paper candidates should ensure that they have been supplied to correct and complete question-paper. No complaint, in this regard, will be entertained after the examination.

Note : Attempt *Five* questions in all, selecting *one* question from each Section. Q. No. **1** is compulsory. All questions carry equal marks.

1. (a) What is linked list ? **2**

(b) Different operation on linked list. 2

(c) Write operations on sets. 2

(d) What is rotation in AVL tree ? 2

(e) Define file and streams. 2

(f) What is Graphs ? 2

(g) Define network flow problem. 2

Section A

2. What do you understand by stack and queue ? Also explain their implementation using linked list. 14

3. What do you understand by quadratic probing ? Also explain hash function. 14

Section B

4. What do you understand by LL, RR, LR and RL ? 14

5. Explain in detail B + tree and 2-3 tree. 14

Section C

6. What do you understand by sets ? Explain different operations on sets along with applications. 14

7. Explain in detail direct file organisation. 14

Section D

8. Explain Dijkstra's Algorithm. 14

9. Explain in detail Floyd Warshall's Algorithm. 14